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Module 6 Introduction
Lesson 1: Numerical Methods and Approximations in Bayesian Computation

Reading: Numerical Methods and Approximations in Bayesian Computation Readings 1h

Video: Numerical Methods and Approximation Pt. 1 4 min

Video: Numerical Methods and Approximation Pt. 2 4 min

Video: Numerical Methods and Approximation Pt. 3 3 min

Practice Assignment: Numerical Methods and Approximations in Bayesian Computation Quiz Submitted

Lesson 2: Simulation Techniques for Bayesian Inference

Lesson 3: Advanced Markov Chain Methods for Bayesian Computation

Module 6 Summative Assessment

Module 6 Summary

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

Numerical Methods and Approximations in Bayesian Computation Quiz

Instructions

Review Learning Objectives

1. Why are logarithms of posterior densities often used in Bayesian computation? 1 / 1 point

- ☒ To avoid computational overflows
- ☐ To better correspond with the maximum likelihood estimate before starting the assignment? I'm here to help.
- ☐ To linearize the data
- ☐ To simplify the computation of integrals

Help me practice

Let's chat

Correct

Correct! Using logarithms of posterior densities helps avoid computational overflows.

Submitted

Dec 6, 2:02 AM IST

Attempts

Unlimited

Retry

2. Which of the following is true about most simulation methods used in Bayesian computation? 1 / 1 point

- ☒ They rely on random sampling
- ☐ They are deterministic algorithms
- ☐ They provide exact values of integration
- ☐ They are analytically derived

Correct

Correct! Simulation methods for obtaining integrals from the desired distribution.

Submitted

Dec 6, 2:02 AM IST

Attempts

Unlimited

Retry

3. Suppose we wish to estimate the integral $\int \sin \theta \cos \theta d\theta$ from 0 to $\sqrt{2}$ Which of the following correspond to the trapezoidal rule? Assume $S = 5$, $\theta^1 = 0$, $\theta^5 = \sqrt{2}$, and θ^i are equidistant. 1 / 1 point

- ☐ $p(\theta|y) = \cos \theta$, $h(\theta) = \sin(\theta)$, $w_0 = \sqrt{2}$, $w_5 = \sqrt{2}$, and $w_s = 2\sqrt{2}$
- ☐ $p(\theta|y) = \sin(\theta) \cos(\theta)$, $h(\theta) = 1$, $w_0 = 1$, $w_5 = 1$, and $w_s = 2$ for all other values of s
- ☒ $p(\theta|y) = 1/\sqrt{2}$, $h(\theta) = \sqrt{2} \sin(\theta) \cos(\theta)$, $w_0 = \sqrt{2}/8$, $w_5 = \sqrt{2}/84$, $w_s = \sqrt{2}/4$ for all other values of s
- ☐ None of the above

Correct

Correct! We have a probability distribution for $p(\theta|y)$. That, combined with the weights yields the approximating corresponding to the trapezoidal rule.